Feng Tian

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RESEARCH INTERESTS

- Computational Imaging
- Computer Vision
- Machine Learning
- Neuroengineering
- Optical Design
- Microfabrication

I work at the intersection of machine learning, computational imaging, and optical design to develop compact imaging and sensing technologies. My research combines deep learning, optimization-based image reconstruction, and inverse design of diffractive optical elements. I develop lensless imaging devices for implantable photography and microscopy applications, with the broader goal of enabling AI-powered 3D sensing in compact devices for machine vision and neuroscience.

EDUCATION AND EMPLOYMENT

University of California, Davis Davis, CA
Postdoctoral Scholar February 2025 - Present

- Engineered integrated physics-informed miniaturized microscope pairing custom doublet microlens array with Multi-Local-FOV ADMM-Net to enable scalable 3D fluorescence imaging (4 mm × 6 mm × 0.6 mm) and single-cell-resolution *in vivo* cortical calcium recording in awake mice.
- Coordinated interdisciplinary research partnerships to translate customized computational miniscopes into active neuroscience paradigms, adapting ultra-compact optical hardware and deep-learning decoders for high-throughput population recording in unconstrained rodent and primate models.
- Developed low-complexity 3D reconstruction architectures utilizing geometrical-optics back-projection and unrolled Richardson-Lucy networks with sparse, partially connected layers, drastically reducing memory overhead and accelerating inference speed.

University of California, Davis Davis, CA
Ph.D. in Electrical and Computer Engineering September 2018—December 2024

- Advanced physics-aware deep learning frameworks for lensless 3D imaging, integrating learnable Fourier-domain models to reconstruct depth-resolved images and through opaque obstacles without experimental point-spread-function (PSF) calibration.
- Designed advanced optical PSF engineering techniques, modeling a double-axicon lens assembly to generate a V-shaped PSF, extending the depth of field to $\sim 350\ \mu$m in wide-field microscopy while preserving lateral resolution.
- Formulated forward optical propagation models and optimization routines to solve ill-posed 3D inverse problems in phase retrieval and wavefront shaping.
- Designed and simulated a specialized multi-photon miniscope optical system, integrating a custom multi-component objective for efficient excitation and collection with an optical grating for temporal focusing beam delivery.

Ams OSRAM Boise, ID
Research Intern June 2022—July 2022

- Conducted design and FDTD simulations using Lumerical and custom Python scripts on metastructure optical elements.
- Developed computational imaging algorithms for object reconstruction through deep learning for

miniaturized imager prototypes.

Nanophotonics and Nanofabrication Group, University of Michigan Ann Arbor, MI
Research Intern May 2018—August 2018

- Conducted proof-of-concept experimental verification of a novel tensorial shear-stress sensor, performing optical excitation and PL spectroscopy on micro-patterned InGaN/GaN nanopillar array samples.
- Developed an autofocusing and mechanical control algorithm for optical microscope.

University of Michigan Ann Arbor, MI
M.S. in Electrical and Computer Engineering September 2016—April 2018

University of Electronic Science and Technology of China Chengdu, Sichuan
B.S. in Electrical and Computer Engineering September 2012—June 2016

- Thesis: All-Optical Modulation Format Conversion Based on Nonlinear Dynamics of Optical Injected Semiconductor Laser

AWARDS

Richard and Joy Dorf Graduate Student Award May 2023
UC Davis Dissertation fellowship April 2023
UC Davis Research Award April 2022
UC Davis Graduate Student Travel Award December 2021

JOURNAL PUBLICATIONS

- **Feng Tian**, Weijian Yang, "Photonics Breakthroughs 2025: Mask-based Integrated Miniscope for Large-scale Volumetric Imaging", ***IEEE Photonics Journal*** 18, 3900408 (2026). [Link]
- **Feng Tian**, Weijian Yang, "Unrolled Richardson-Lucy deconvolution network with partially connected layers in computational microscopy", ***Optics Express*** 34, 18676-18689 (2026). [Link]
- **Feng Tian**, Ben Mattison, Weijian Yang, "DeepInMiniscope: Deep learning–powered physics-informed integrated miniscope", ***Science Advances*** 11, eadr6687 (2025). [Link]
- Yunyang Li, Zixiao Zhang, **Feng Tian**, Yryx Y. Luna-Palacios, Israel Rocha-Mendoza, Weijian Yang, "V-shaped PSF for 3D imaging over an extended depth of field in wide-field microscopy", ***Optics Letters*** 50, 383-386 (2025). [Link]
- **Feng Tian**, Weijian Yang, "Learned lensless 3D camera", ***Optics Express*** 30, 34479-34496 (2022). [Link]
- **Feng Tian**, Junjie Hu, Weijian Yang, "GEOMScope: Large field-of-view 3D lensless microscopy with low computational complexity", ***Laser & Photonics Reviews*** 15, 2100072 (2021). [Link]
- Jingyang Sui, Kunook Chung, **Feng Tian**, Pei-Cheng Ku, "A tensorial shear stress sensor based on light-emitting GaN nanopillars", ***Applied Physics Letters*** 115, 021103 (2019). [Link]

CONFERENCES PROCEEDINGS

- Feng Tian, Weijian Yang, Ben Mattison "Physics-aware lensless miniscope for neuronal imaging (Invited Paper)", Proc. SPIE PC13333, Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences II (2026).
- Feng Tian, Weijian Yang, "Compact 3D fluorescence microscopes using a single layer of microlens array (Invited Paper)", Proc. SPIE PC13330, High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond X (2025).
- Yunyang Li, Shu Guo, Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, "High-speed short-line-excitation two-photon microscopy to image brain activity (Invited Paper)", Proc. SPIE PC13303, Neural

Imaging and Sensing (2025).

- Feng Tian, Weijian Yang, “Physics aware lensless miniscope for neuronal imaging”, Proc. SPIE PC13333, Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences II (2025).
- Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, “High-speed miniaturized two-photon microscopy with elliptical beam excitation”, Proc. SPIE PC12828, Neural Imaging and Sensing 2024 (2024).
- Feng Tian, Weijian Yang “Miniaturized and Thin Microscope with Learnable Real-Time Reconstruction”, Conference on Lasers and Electro-Optics, Optica Publishing Group, ATu3Q.3 (2023).
- Yunyang Li, Zixiao Zhang, Feng Tian, Yryx Luna-Palacios, Israel Rocha-Mendoza, Weijian Yang, “V-Shape PSF for 3D Wide Field Microscopy”, Conference on Lasers and Electro-Optics, Optica Publishing Group, ATu4Q.3 (2023).
- Ben Mattison, Shing-Jiuan Liu, Feng Tian, Weijian Yang, “High Speed Miniaturized Multiphoton Microscopy With Elliptical Beam Excitation”, Biophotonics Congress: Optics in the Life Science, Optica Publishing Group, JTu4B.33 (2023).
- Feng Tian, Weijian Yang, “Miniaturized and Thin 3D Microscope With Learnable Real-Time Reconstruction”, Biophotonics Congress: Optics in the Life Science, Optica Publishing Group, BM3B.3 (Student Paper Finalist) (2023).
- Feng Tian, Weijian Yang, “Miniaturized fluorescence 3D microscope with deep learning reconstruction”, Proc. SPIE 12385, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXX (2023).
- Feng Tian, Weijian Yang, “Computational 3D imaging with micro optics and data-driven deep learning reconstruction”, Proc. SPIE 12019, AI and Optical Data Sciences III (2022).
- Feng Tian, Weijian Yang, “3D Camera using a Single Layer of Microlens Array and a Learned Reconstruction Algorithm”, Conference on Lasers and Electro-Optics, Optica Publishing Group, JW3A.7 (2022).
- Feng Tian and Weijian Yang, "Miniaturized and Thin 3D Microscope with Fast Reconstruction," Conference on Lasers and Electro-Optics, Optica Publishing Group, SM5O.3 (2022).
- Feng Tian, Junjie Hu, and Weijian Yang, “Large field-of-view 3D Imaging using Random Microlenses”, Conference on Lasers and Electro-Optics, Optica Publishing Group, SW2D.3 (2021).
- Feng Tian, Junjie Hu, and Weijian Yang “Large field-of-view three-dimensional imaging with low computational complexity using random microlens array”, Proc. SPIE 11649, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXVIII (2021)
- Feng Tian, Junjie Hu, and Weijian Yang, “Three-dimensional Imaging with a Single Layer of Random Microlens Array”, Conference on Lasers and Electro-Optics, Optica Publishing Group, STh3M.3 (2020).
- Kunook Chung, Feng Tian, Jingyang Sui and Pei-Cheng Ku, “Tensorial Shear Stress Sensing Using Elliptically-Shaped Nanopillar Light-Emitting Diodes”, Conference on Lasers and Electro-Optics, Optica Publishing Group, AM4I.4 (2019).

TEACHING EXPERIENCE

University of California, Davis — Davis, CA

Teaching Assistant

EEC 100 – Circuits II	Winter 2024
EEC 100 – Circuits II	Fall 2023
EEC 130A – Electromagnetics I	Fall 2022
EEC 243 – Optical Imaging and Microscopy	Winter 2021
EEC 140A – Principles of Device Physics I	Fall 2019
EEC 140B – Principles of Device Physics II	Spring 2019

PROFESSIONAL SERVICES

Journal Reviewer

- Laser & Photonics Reviews
- Photonics Research
- Optics Express
- Optics Letters
- IEEE Transactions on Signal Processing
- Applied Physics Letters
- Applied Optics
- Frontiers in Physics